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A Note on Microeconomics for Strategists

One of the fundamental units of analysis for the strategist is the industry or market in which a firm competes. A market consists of a set of firms vying to sell similar goods to a similar set of customers. Economists' models of markets are simplified to allow rigorous analysis and only roughly approximate real markets. Yet microeconomic models incorporate concepts that are invaluable to the business strategist, and they provide a foundation on which to build structured analyses of real markets.

This note summarizes the core ideas about the microeconomics of markets that are most relevant to business strategy. Sections I and II develop two basic building blocks of any market, demand and supply. Section III discusses how demand and supply interact to determine the quantity of goods traded in a market and the price paid for those goods, with special attention to the way that external events influence the quantity traded and the price paid. Section IV presents the important benchmark of "perfect competition," in which equally matched firms compete so vigorously and market entry is so easy that no firm earns more than its cost of capital. Section V explores the ways that real markets depart from perfect competition. These departures lie at the heart of long-run profitability.

I. Demand Analysis

Buyers, be they consumers or businesses, want products and services. Consumers buy goods because they derive some utility or happiness from their purchases. Businesses purchase inputs to help generate a profit. In either case, a buyer's *willingness to pay* for a particular unit of a particular good is defined as the highest price at which the buyer would choose to purchase the unit of the good. Willingness to pay is *not* necessarily what the buyer thinks is a fair price or what the buyer wishes to pay. Rather, it is the price at which the buyer is just indifferent between purchasing the good and not purchasing it. The buyer always prefers a lower price, but will pay any price up to his or her willingness to pay.

A buyer's willingness to pay is influenced by a number of factors:

- The buyer's *tastes or needs* determine the desire of the buyer to purchase a good relative to competing ones. As tastes and needs for a particular good change, the willingness to pay for the good will rise or fall.

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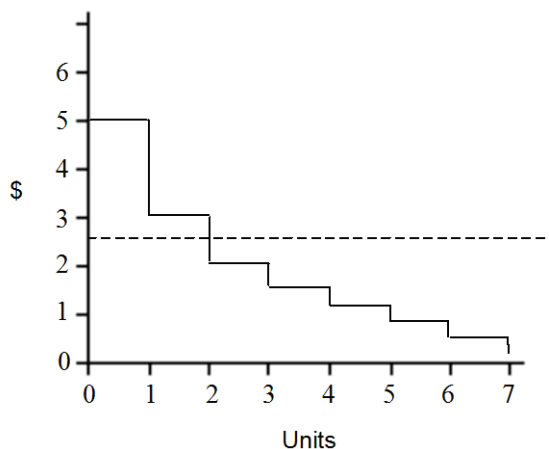
- A consumer's *income or wealth* or, in the case of a business buyer, the buyer's financial resources, influence the buyer's willingness to pay for a particular good. The willingness to pay for most goods increases as buyers become wealthier, but for some goods (e.g., potatoes), it may actually fall. In industries where buyers' income or wealth affects willingness to pay a lot (e.g., the market for yachts), demand tends to be highly cyclical.

- Prices and availability of *substitute goods* also affect consumers' willingness to pay for a good. Substitutes are goods that can perform the same function as the good in question. When prices in the steel industry are very low, for instance, the demand for aluminum, a steel substitute, tends to fall.

- Similarly, *complementary goods* can affect willingness to pay for a good. Two goods are complements if a buyer is willing to pay more for one when he or she already owns the other, and vice versa. For example, widespread availability of inexpensive compact discs boosted the willingness of many consumers to pay for CD players in the 1980s.

A *demand curve* for an individual buyer summarizes the buyer's willingness to pay for various quantities of a good. Equivalently, it specifies the quantity of a good that the individual would purchase at each price. **Figure 1** shows a typical demand curve for a single buyer. The curve shows that the buyer is willing to pay \$5 for the first unit of the good, \$3 for the second unit, \$2 for the third, and so forth. If offered a price of \$2.50 per unit, this buyer would purchase two units.

Figure 1 Demand Curve for an Individual Buyer



A demand curve usually *slopes downward* since buyers are normally willing to pay less and less for additional units of a good as they buy more and more. Your first doughnut of the morning may be very valuable to you, the second less so, and the tenth far less so. For business buyers, the downward slope of the demand curve reflects decreases in the productivity of additional purchases. While the first network server installed to provide e-mail to employees might be very valuable, second and subsequent servers might prove merely redundant. In this case, the firm would have a much lower willingness to pay for additional computers than for the first.

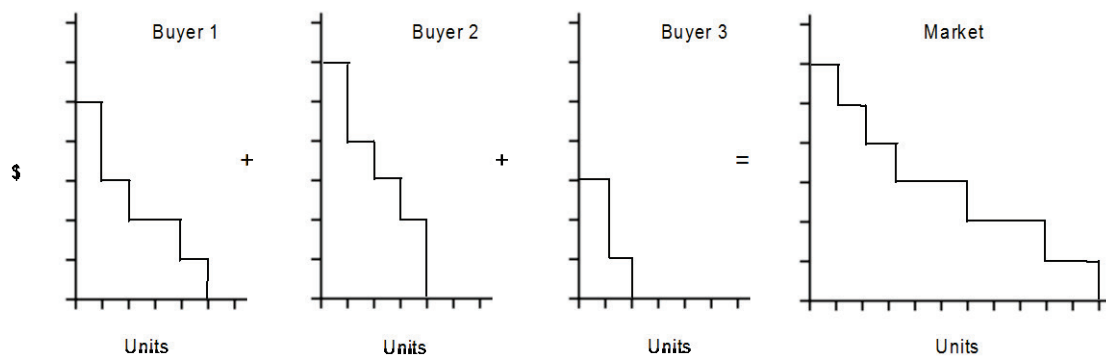
Shifts in the Demand Curve. Changes in the factors discussed above—a buyer's tastes and needs, his or her income or wealth, and conditions in the markets for substitutes and complements—alter willingness to pay and hence shift the buyer's demand curve up or down. An unexpectedly large year-end bonus, for instance, might boost a consumer's demand for gourmet chocolates. The bonus would cause the consumer's demand curve to move rightward: at any given price, he or she would want to buy more chocolate. Understanding the determinants of demand is important in anticipating

how industry prices and quantities respond to changes in buyer or seller behavior, a subject discussed further in section III.

Market Demand

The sum of the demand of individual consumers or businesses at each price results in a *market demand curve* (Figure 2). The market demand curve reports, at any given price, the aggregate quantity demanded. All of the factors that shift individual demand curves up or down also reposition the market demand curve.

Figure 2 Market Demand Curve Constructed from Individual Demand Curves



One can think of the market demand curve as *arraying individual buyers in order of their willingness to pay*. Buyers who have an ample income or a high need for the good constitute the upper left end of the curve. Buyers with tighter budgets or less interest in the good comprise the lower end of the curve. In the market for personal laser printers, for example, the upper end of the demand curve might represent high-income professionals who maintain home offices. The next portion might include individuals who are well-off, but do not use their home computers for particularly important correspondence and therefore find ink jet printers more acceptable substitutes. The lower end of the demand curve might include cash-strapped students. Ultimately, the tail end of the demand curve would be made up of people who do not own computers and therefore are not willing to pay anything for a printer. Note that even when each individual buyer desires only a single unit of a good, the market demand curve is still downward-sloping since it arranges buyers according to their willingness to pay.

Strictly speaking, it makes sense to sum up individual demand curves only for *identical* goods. In a few markets, goods really are identical for all intents and purposes (e.g., the markets for certain chemical feedstocks). Other kinds of products, however, differ in myriad ways, and sometimes no two sellers produce goods that can reasonably be called identical. (Consider, for instance, the work of visual artists.) In addition, real buyers differ in exactly what types of goods they prefer. It is therefore often useful to group similar products together somewhat loosely in constructing market demand curves. An important skill in utilizing demand analysis is determining just how broad a market can be fruitfully analyzed as a single entity.

Demand Segments and Price Discrimination

Though it is often useful to talk about demand in a broad product category, it is important to keep in mind that demand is likely to be composed of many distinct segments. In the market for laser

printers, for instance, we noted that there are distinct clusters of customers. These segments differ in their willingness to pay not only for a specific type of printer, but also for speed, additional fonts, extra memory, and other features. Successful product development and marketing initiatives require a deep understanding of the segments comprising this market demand curve.

The fact that buyers or subgroups of buyers differ in their willingness to pay raises an interesting dilemma for a seller. At any price that entices most buyers to purchase the good, the buyers with the very highest willingness to pay are getting a terrific deal: they are purchasing the product for far less than they would be willing to pay. Firms often try to close this gap by charging different prices to different buyers, a practice known as *price discrimination*.¹

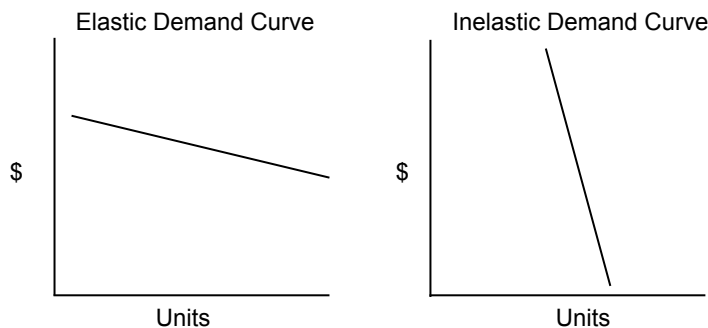
Ideally, a seller would like to charge each consumer exactly his willingness to pay for each unit of the good, but this is almost always impossible. Firms employ a number of clever tactics to achieve similar ends, however. In the laser printer market, for example, IBM once offered two models that were identical in every way except one: one had an additional chip that slowed the print speed by half.² The faster model, built at essentially the same cost, sold at a significant premium. The company realized that the market demand curve for printers consisted of at least two segments: one with higher willingness to pay for speed and one with lower willingness to pay. IBM found a way to serve both segments and to charge a higher price to the higher-value user group.

Price Sensitivity, or Elasticity, of Demand

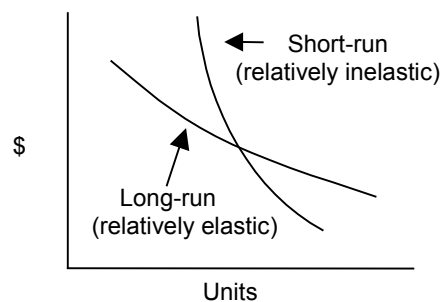
The slope, or steepness, of a market demand curve measures how responsive buyers are to changes in price, and is crucial to the way markets adjust. For some goods, the curve is flat or near-flat (**Figure 3**): the quantity that buyers want is highly sensitive to the price charged, and a small dip in price sparks a large surge in the quantity demanded. We say that such a good has a high *price elasticity of demand* because demand stretches a great deal as price changes. This is typical of goods such as butter that have close substitutes (e.g., margarine). An inelastic demand curve is steeper (see **Figure 3**), indicating that consumers buy nearly the same amount of the good even if its price rises or falls significantly. Necessities with few substitutes such as water tend to have inelastic demand curves.

¹ Laws like the U.S.'s Robinson-Patman Act may prohibit price discrimination when a firm sells *identical* goods to multiple consumer groups.

² Raymond J. Deneckere and R. Preston McAfee, "Damaged Goods," *Journal of Economics and Management Strategy* (5), Summer 1996, pp. 149-74.

Figure 3 Price Elasticity of Demand

The price elasticity of demand is usually a function of the *time period* under consideration. In the near term, an increase in the price of a product may have little effect on demand because it takes time to develop substitutes and learn how to use them. Over time, however, substitutes can often be found and used. Demand for petroleum, for instance, is more sensitive to price over longer time horizons because of the time it takes to develop alternative fuel sources and promote conservation. Indeed, for most goods, demand is more responsive to price—that is, the demand curve is flatter or more elastic—in the long run (**Figure 4**).

Figure 4 Short-run and Long-run Demand Curves

A thorough grasp of the nature of demand for a good is important to the strategist. As we will see below, it aids him or her in understanding how a price change or a shift in a related market affects aggregate output. It also helps the strategist see how such changes alter the mix of buyers who are actively purchasing a product. Analysis of the segments that make up the demand curve can be helpful in understanding how to expand the market. One might, for instance, design special products or pricing schemes to bring buyers with lower willingness to pay into the market, as American Express did with its special credit card for students. Conversely, one might offer a high-end product like American Express' platinum card in order to boost the profitability of the higher willingness to pay segments.

In economists' models, demand curves are generated by forces beyond the control of firms. In real markets, firms take actions to influence the size of demand and the shape of the demand curve. Nike's advertising has greatly expanded the demand for athletic shoes and has perhaps made consumers less price-sensitive. Sales efforts by pharmaceutical makers shape demand curves for drugs by communicating their safety and efficacy to prescribing physicians. Microsoft has invested in Internet-related technologies—complementary products—in part to boost demand for its Windows operating system.

II. Supply Analysis

Just as a demand curve shows how much of a good a buyer will purchase at each price, a supply curve describes how much of a good a supplier will choose to provide at each price. In this section, we consider the supply decisions of firms that take market prices as given. That is, each firm is small enough relative to its entire industry that its decisions do not affect the market price. When firms are large relative to the market or, in the extreme, enjoy a monopoly, they may try to influence market prices through their supply decisions (see Section V). In examining supply decisions, we focus on firms that maximize profits, rather than some other objective like market share.

A firm's supply decision really consists of two decisions. The first decision is whether to compete in the market at all. We call any time horizon long enough to permit entry and exit decisions the *long run*. The second decision is the *short-run* choice of how much to produce in light of current market conditions, *given that the firm is in the market*. Note that under this definition the short run may in fact be as long as several years—in industries where capacity additions require long planning and construction cycles, for example. It is easiest to explore the short-run supply decision first, then consider the long run.

Supply in the Short Run

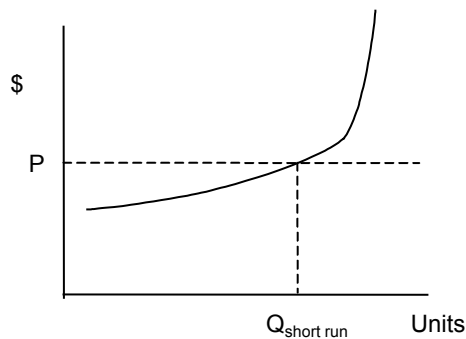
A firm's decision about how much to supply in the short run starts with its costs. Costs are generated by the various activities of the firm, ranging from manufacturing and sales to distribution and administration. Different activities generate costs with different attributes.

The most important distinction is between *fixed costs* and *marginal costs*. Fixed costs are expenses that must be paid if a firm is open for business, regardless of the volume of goods produced. For instance, a paper maker must incur certain administrative costs, rent or depreciation on buildings and equipment, salaries of managers, and so forth, no matter how much paper it produces. Marginal costs are costs that vary substantially with the level of production: the cost of raw materials such as pulp and chemicals, some direct labor costs, some repair and maintenance expenditures, etc. Producing incremental units of output leads to incremental increases in such costs.

In the short run, fixed costs are unavoidable, and they should not influence immediate choices. Only marginal costs should affect short-run supply decisions. Managers often make poor decisions because they fail to grasp this basic point. Many capital-intensive firms have lost market share and profits because they set price to "cover fixed costs" during industry downturns.

In considering marginal costs, it is important to include both *cash costs* and *opportunity costs*. To produce more fine paper, a paper maker must incur cash expenditures for additional pulp. Such expenditures are part of the firm's marginal *cash costs*. However, the paper company may also have to divert scarce capacity away from other products to expand fine paper supply, forgoing the opportunity to make profit on those products. Marginal costs include not only cash costs but also any non-cash *opportunity costs* of forgone alternatives. Sophisticated firms consider total marginal costs, not just cash costs, when making supply decisions.

At any level of production, the firm faces a certain marginal cost of producing an additional unit of output. The plot of marginal cost at each level of output gives a *marginal cost curve* (**Figure 5**). A firm's marginal cost curve typically slopes upward since it becomes increasingly costly to squeeze incremental output out of busy production lines and over-stretched personnel as the firm produces more. As a firm reaches 100% capacity utilization, the marginal cost curve becomes a nearly vertical line.

Figure 5 Marginal Cost Curve of a Firm

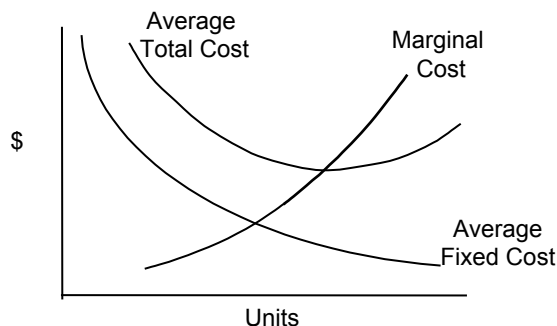
At market price P in **Figure 5**, how much should the firm produce? For any level of output below $Q_{\text{short run}}$, an incremental unit will generate revenue P and will cost, on the margin, something less than P . Hence the firm should continue to produce more and more until its output is $Q_{\text{short run}}$ and its marginal costs rise to P . Looking at **Figure 5** in a different way, then, the marginal cost curve summarizes a firm's short-run supply decision: it shows how much a price-taking firm will produce at any price P . For this reason, the marginal cost curve is precisely the same as the *short-run supply curve* of an individual firm.

Supply in the Long Run: Entry and Exit

While marginal costs are paramount in the short run, fixed costs are pivotal in the long-run decision to enter or leave a market. In the short run, a firm should expand production so long as the price it receives meets or exceeds its marginal costs. Any payment received above those costs makes some contribution toward covering the fixed costs that are unavoidable in the short run. In the long run, however, fixed costs are avoidable, and a firm will generally stay in a market only if it expects to cover its *fixed costs* as well as its marginal costs.

A firm's *fixed costs* include both out-of-pocket expenditures that do not vary with production—including rent, for example—and the opportunity cost of the capital the firm employs in its business. If it were not tied up in the business, that capital could be profitably invested elsewhere. Consequently, a firm's fixed costs must include the sum that could be earned by putting the capital to alternative uses. When we say that a firm earns no profits, or breaks even, we mean that the firm earns exactly its cost of capital. This is clearly not the same as an accounting measure of profitability like net income, which may well be positive even if the firm fails to earn its cost of capital. Many firms earn an accounting profit but still fail to recoup their cost of capital.

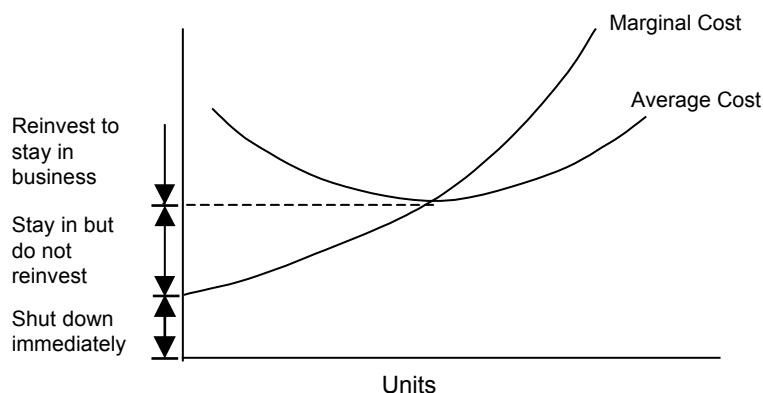
A firm's *average cost* includes both its fixed and its marginal costs. Specifically, the average cost curve shows, at each level of output, the fixed costs of the firm and the total marginal costs divided by the number of units produced. The typical average cost curve has a distinctive "U" shape due to the opposing effects of fixed and marginal costs. The average fixed cost per unit declines with production since fixed costs are spread over more and more units. Eventually, though, as a firm expands and its facilities and personnel are stretched thin, marginal costs rise so high that average total costs begin to rise again. **Figure 6** shows a typical set of average cost curves.

Figure 6 Marginal and Average Cost Curves of a Firm

The average cost curve bottoms out at some level of output. This is the scale that allows the firm to produce as efficiently as possible. As long as the firm expects the market price to exceed this minimum average cost, the firm will choose to enter and stay in the business. Once it has entered, the marginal cost curve will dictate its short-run supply decisions.

The discussion so far has been framed in terms of an entry decision. *Exit decisions* are very similar. Consider a firm in a market with declining prices (**Figure 7**). So long as the price remains above the minimum of its average cost curve, economic theory says that the firm should continue to reinvest in fixed assets as necessary to stay in business. If the price drops below the minimum of the average cost curve, the firm may remain in business in the short run, as long as prices cover marginal costs. It should, however, exit rather than pay fixed costs for subsequent periods. If price drops below the minimum of the marginal cost curve, the firm should simply shut down immediately. Real exit decisions are messier than this, of course, but the economic logic is similar.

In sum, a firm should enter and remain in a business so long as it expects the market price to be greater than its average total costs. Once in business, the firm should expand output until the marginal cost of an incremental unit rises to the market price.

Figure 7 Exit Decisions

Shifts in the Supply Curve. Two factors have a particularly large influence on the position and shape of a supply curve: the costs of inputs and technology. When the cost of a marginal input rises or falls, so does the marginal cost curve. A firm then produces less at any given price or, equivalently, prices must be higher to induce a given amount of output. A weather problem that hurts the coffee crop and raises bean prices will shift the supply curve of a coffee processor upward.

Similarly, a fall in the price of copper will lower the supply curve of a manufacturer of copper wire, implying that more wire is supplied at any given price.

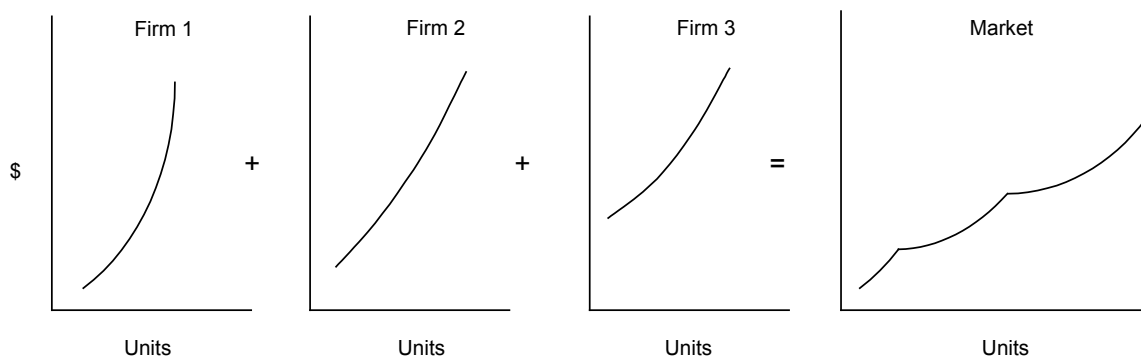
When the cost of a *fixed* input rises, so does the average cost curve. Firms will then reassess their long-run decisions to participate in the market. Capital expenditures necessitated by, say, the Clean Air Act may cause some oil refiners to exit the business.

Technological change can move costs downward if more efficient ways are discovered to produce a good. Consider, for instance, a firm that produces and distributes music. In order to sell an incremental compact disc, the firm must purchase a blank disc, burn music onto it, and package and distribute its product—all costly steps. With the advent of the Internet and of CD players that record music, the firm can now deliver music by downloading it to the customer's computer. The marginal and fixed costs associated with this process are much lower than those incurred by the traditional distribution route. Indeed, the marginal costs are nearly zero. In this case, a change in technology might cause a dramatic shift in the supply curve.

Market Supply

The *market supply curve* is obtained by summing up the quantities that individual firms will produce at each price (**Figure 8**). The industry supply curve slopes upward for two reasons: the supply curves of individual firms slope upward, and firms differ in their productive efficiencies. As price increases, not only do existing firms and plants expand output, but less efficient firms and plants are brought on stream.

Figure 8 Market Supply Curve Constructed from Individual Firm Supply Curves



It is often illuminating to think of the market supply curve as arraying individual firms or plants from the most efficient to the least efficient. At low prices, only the most efficient producers continue to operate, and the quantity supplied is relatively low. These efficient firms make up the lower left end of the supply curve. As prices rise, incumbent producers may expand facilities or add new capacity. Some additional firms whose cost positions did not permit them to be profitable at lower prices may also enter the industry. These less efficient firms make up the upper right end of the supply curve.

Price Sensitivity, or Elasticity, of Supply

An important aspect of market supply is its *elasticity*, which is analogous to the notion of the elasticity of demand. Where the *price elasticity of supply* is high, the supply curve is nearly flat and a small increase in price evokes a lot of new output. This might arise in industries where facilities used for other things can be readily converted to producing the good in question, where capacity can be

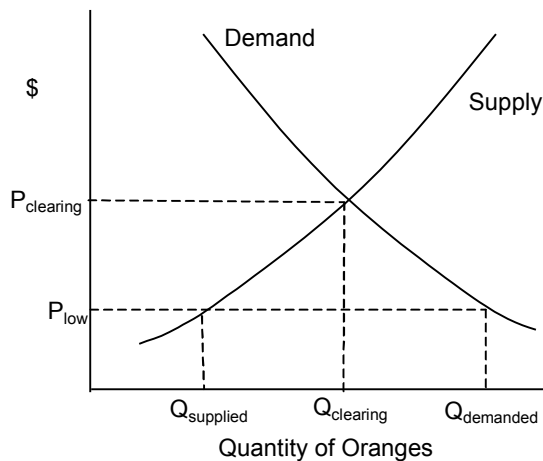
expanded easily, or where new firms can enter the business readily. Other industries have a low price elasticity of supply: the supply curve is steep and an increase in price brings on very little additional output. Consider, for instance, what happens when a sporting event such as the Super Bowl comes to a city. Hotel room rates rise, but the supply of rooms does not expand much in response. In contrast, if the price of t-shirts emblazoned with a team logo rises, shirt makers rush to expand output. The supply of team t-shirts is far more elastic than the supply of hotel rooms.

A supply curve can be more elastic in one region than in another. The curve typically becomes inelastic (that is, steep) as firms in the industry run into capacity constraints. Also like the demand curve, the supply curve tends to be more elastic in the long run than in the short run. Suppose that steel prices increase. In the short run, existing steel makers are typically limited in the amount they can supply and new firms often cannot enter quickly. As a result, total steel supply does not respond a great deal to an increase in price. In the long run, however, firms can add capacity and entrants can establish themselves. These factors make the long-run supply curve for steel relatively elastic (that is, flat). The same reasoning holds true for a price decrease. In the short run, barriers to exit and shrinkage prevent inefficient firms from leaving the steel market and efficient firms from shrinking; output adjusts little. In the long run, however, poor steel makers drop out of the market and good producers scale back; output adjusts more.

An analysis of supply is useful to the strategist. A thorough understanding of supply allows a set of managers to predict how individual firms and the market as a whole will react to shifts in input costs and technology. A grasp of the elasticity of the industry supply curve allows one to anticipate how responsive the quantities supplied will be to prices and how quickly prices will rise or fall in response to changes in demand. Also valuable is an understanding of the marginal cost curves of individual firms. As we will see soon, firms with high marginal cost curves will typically be the ones squeezed out of a market when the market falls into a slump, especially in commodity markets.

III. Market Equilibrium

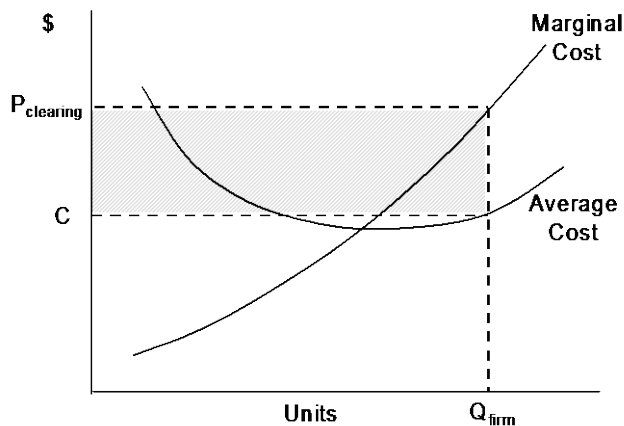
The *market equilibrium* in an industry is determined by overlaying the industry demand and supply curves. **Figure 9** shows the demand and supply curves for oranges. At price P_{low} , supermarkets, orange juice makers, and other buyers are eager to purchase Q_{demanded} oranges, but orange growers are willing to provide only Q_{supplied} . With *excess demand*, buyers scramble to purchase the few oranges that are available, and prices get bid up. Over time, growers add to their crop, new growers enter the market, and buyers turn to substitutes such as tangerines. Eventually, the market comes to the price P_{clearing} and output Q_{clearing} , at which supply and demand are in balance. We call this point the market equilibrium and say that the “market clears” at this price and quantity.

Figure 9 Equilibrium in the Market for Oranges

Alternatively, if the price is initially higher than P_{clearing} , orange growers supply more fruit than buyers will purchase. Growers find they cannot sell all of their oranges and inventories pile up. Prices fall as growers try to rid themselves of *excess supply* before the inventory goes bad. They reduce their plans for future harvests, and perhaps some growers leave the market. As prices fall, buyers are willing to buy more and more oranges. The adjustment continues until supply and demand are again in balance at the equilibrium price and quantity.

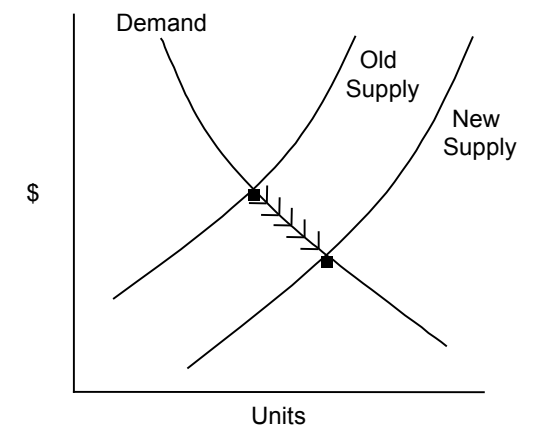
Real markets rarely rest precisely at equilibrium. Because supply and demand curves do not stay perfectly still for long and prices and quantities take some time to adjust, there is usually some excess supply or excess demand at any point in time. However, actual prices and quantities do move toward the equilibrium and, in competitive markets, rarely depart a great deal from equilibrium for a long time. In hotly contested arenas such as global currency markets, adjustment to equilibrium can be nearly instantaneous.

Once one knows the market price, one can calculate each firm's profit from its cost curves. **Figure 10** plots the marginal and average cost curves of a particular firm. Given a market price P_{clearing} determined by the industry-wide equilibrium, the firm will produce a quantity Q_{firm} . In doing so, it incurs an average cost C . The firm's profit is simply the difference between the price and cost per unit multiplied by the number of units it sells – the shaded region in **Figure 10**.

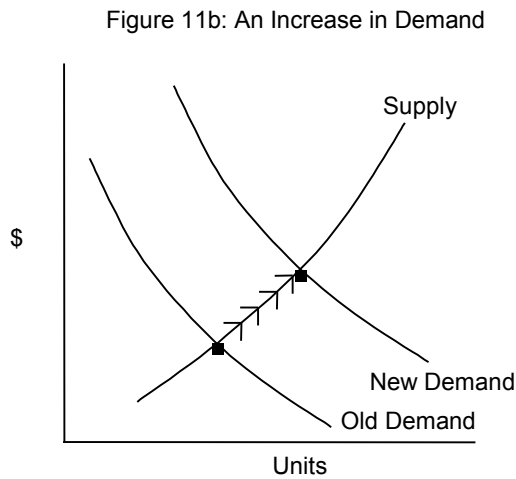
Figure 10 Supplier Profit at an Equilibrium

Market Adjustment

When supply or demand shifts, the market equilibrium moves in predictable ways. For instance, an increase in supply, *i.e.*, a rightward movement of the supply curve, causes the equilibrium price to fall and the equilibrium quantity to rise (**Figure 11a**). In the market for personal computers, for example, rapid declines in component prices have caused the supply curve to drop precipitously. Accordingly, computer prices have plummeted and volumes have soared. An increase in demand, *i.e.*, a rightward shift of the demand curve, causes both price and quantity to rise (**Figure 11b**). This occurred in the market for diamonds in the early 1990s as rising wealth, especially in Asia, caused a surge in demand.³

Figure 11a An Increase in Supply

³ Note that an increase in supply or demand shifts the curve rightward, meaning that a higher quantity is supplied or demanded at each price. It is important not to be confused by the fact that *increases* in supply shift the supply curve *down*, while increases in demand shift the demand curve *up*.

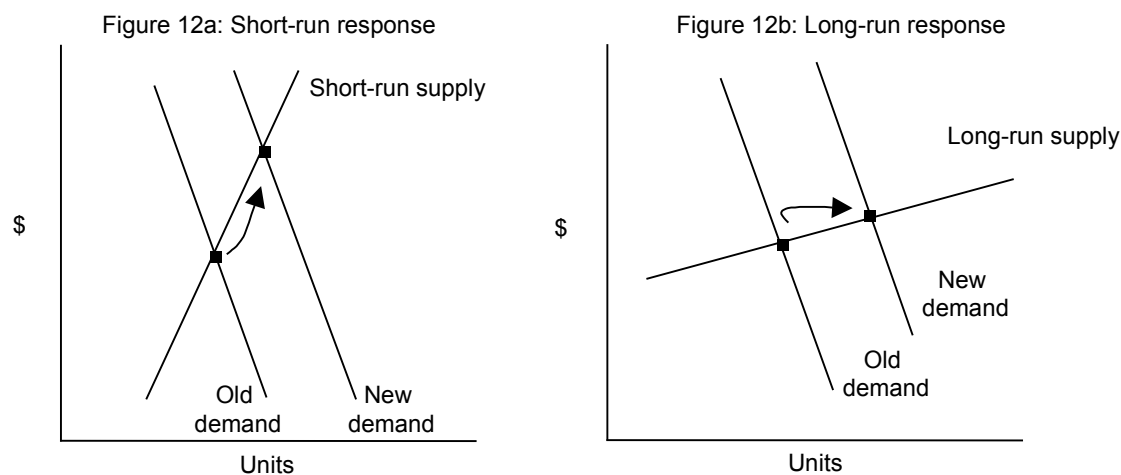
Figure 11b An increase in Demand

With a basic understanding of market adjustment, a strategist can not only predict the consequences of changes in supply or demand, but can also make better sense of changes in prices or output. Consider, for instance, a firm that is considering entry into the trucking business. Managers in the firm observe that trucking prices have increased recently. Does this make entry more or less attractive? The answer may hinge on whether the price increase is due to a surge in demand or a contraction of supply (for instance, because of cost increases). If the managers see that trucking volumes have increased at the same time that prices have risen, they can safely conclude that demand must have grown. If volumes have declined as prices have risen, then a contraction in supply is likely.

If one knows the elasticities of supply and demand, one can make even more informed predictions about the consequences of a particular change in supply or demand. One can say, for instance, whether the adjustment is likely to take the form primarily of a price change or of a change in output. Markets with inelastic supply and demand (steep curves) tend to adjust principally by means of price changes. Supply and demand shifts in industries with elastic supply and demand (flat curves) tend to cause output fluctuations. Recall that both supply and demand tend to be relatively inelastic in the short run but more elastic in the long run. Accordingly, markets often respond to changes by means of price adjustments in the short run and output changes in the long run.

Consider, for instance, the market for apartments in Allston, Massachusetts, the home of Harvard Business School. The supply of apartments is relatively fixed and inelastic in this crowded, urban area; the supply curve is close to vertical. If the number of students admitted to HBS increases, the demand curve shifts out since more apartments are demanded at every price. Because the short-run supply curve is steep, adjustment to the increase in demand takes the form, primarily, of a price change: rents increase (see **Figure 12a**). The number of apartments rented hardly changes.

Over time, the increase in rents encourages additions to the housing stock. Homeowners might convert unused attic space into student apartments, and developers might build new apartment buildings on vacant lots. For this reason, the long-run supply curve is flatter than the short-run supply curve—increases in price *do* lead to increases in the quantity supplied eventually. As a result, in the long run the effect of an increase in demand is seen more in increases in the quantity of apartments rented and less in increases in rents (see **Figure 12b**).

Figure 12 Price and quantity response to an increase in demand

IV. Perfect Competition

Three assumptions about the nature of the market underlie the discussion of market equilibrium above. Economists expect markets to tend toward the prices and quantities predicted by the model of market equilibrium when these conditions hold:

Products are identical. Sellers offer indistinguishable products, and buyers are equally willing to purchase from all sellers. Without this assumption, it would not make sense to aggregate curves and represent them on a single graph.

- **Many, small participants (price-taking).** There are numerous suppliers and buyers, and no single firm or consumer accounts for a large portion of production or consumption. Consequently, individual suppliers and buyers know that their decisions do not affect the market price and can realistically believe that the price is given by forces beyond their control. Suppliers and buyers act as *price-takers*, not price-makers. Accordingly, sellers and buyers cannot engage in strategic manipulation of supply and demand. A seller, for instance, cannot threaten to reduce output in order to incite a bidding war among buyers; buyers would simply turn to other sellers. This price-taking assumption rules out not only large, influential players, but also cartels that allow small players to act in unison.

Under the assumption of price-taking, one can simply read the quantities demanded and supplied off the willingness-to-pay and marginal cost curves at the market price.

Full information. Buyers and sellers are fully informed about the products and prices available in the market. There is, for instance, no uncertainty about the market price or about the quality of a good or its usefulness to the buyer.

Together, these assumptions generate the market equilibrium discussed in the last section. As **Figure 10** showed, a firm may earn a profit in a market equilibrium if the market-clearing price is greater than its average cost. Such a situation should, however, catch the attention of other firms already in the business and of potential entrants. Eager to earn similar profits, those others will build additional capacity or enter the industry if they can also produce the good at an average cost less than the market price. This drives down prices and industry profits. If existing rivals and entrants can

imitate the operations of the profitable firm entirely, such competition and entry will completely dissipate profits.

The idea that competition and imitation tend to erode profits is captured by economists' models of *perfect competition*. In addition to identical products, numerous small participants, and full information, perfectly competitive markets have:

Identical sellers. Supplying firms have access to the same production technology and inputs as one another, and on the same terms as one another.

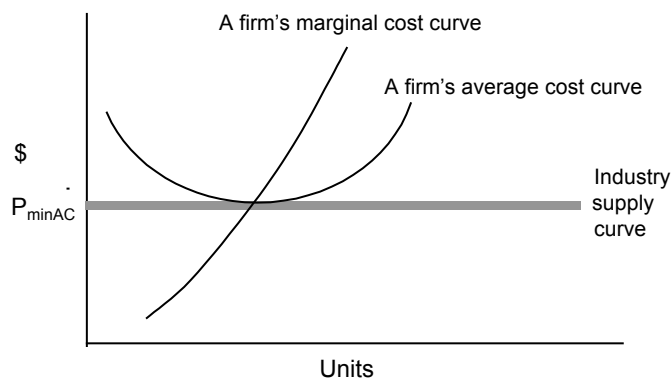
Free entry. New firms can enter the market on the very same terms as existing firms. They may have to pay a fixed cost in order to set up in the market, but the cost they pay is no higher than that of firms already in the market. Moreover, there are many, many potential entrants available to enter the market if it is profitable.

Free exit. Similarly, firms can exit without incurring extra costs. While firms may or may not be able to recoup the costs of their assets (by selling their equipment, for example), they do not face any new costs as a result of the decision to shut down.

These additional conditions lead to a lack of friction in the long-run adjustment of a market. The first condition implies that the best technologies and best practices can be imitated costlessly by all firms. Thus, all firms in the market are equally efficient. The second extends this frictionless imitation to new firms as well as existing firms. The third ensures that firms never get "stuck" in a loss-making industry.

The competition that ensues in such a market is intense. So long as any supplier is earning a profit above its cost of capital, more suppliers rush into the market. By charging lower prices for identical products, they immediately attract customers, who are fully informed of the lower prices. Prices drop until no additional seller is tempted to enter; the marginal firm sees the market as an arena in which it can earn nothing more than its cost of capital. Moreover, since suppliers are identical, the same is true of all the firms actually in the market: none earns anything more than the competitive cost of capital.

We can characterize a perfectly competitive market in terms of the concepts presented earlier in this note. Because a large number of entrants stand ready to enter the market at the slightest scent of a profit, the industry supply curve is a *horizontal line at the very bottom of the representative firm's average cost curve* (**Figure 13**). If price pushes even a hair above this level, firms will be able to make a profit and entrants will rush in, driving prices back down and dissipating industry profits. Thus, free entry ensures that firms do not make positive profits. Similarly, if price drops below this level, firms in the industry will make negative profits. Since exit is free, firms will exit rather than make losses, driving prices back up so that remaining firms break even.

Figure 13 Industry Supply Under Perfect Competition

Together, free entry and free exit imply that firms earn zero profits (exactly their cost of capital) and that the price is exactly $P_{\min AC}$ regardless of the level of demand. As industry demand rises or falls, firms enter or exit, but prices do not change; this implies that the *industry* supply curve is horizontal.

A natural question is, what's so perfect about this form of competition? From the manager's point of view, it is hard to imagine anything further from ideal. An appendix discusses how, from the perspective of the total well-being of society rather than the point of view of a single firm, perfectly competitive markets have some very attractive features.

V. Departures from Perfect Competition

While the forces of competition, imitation, and entry certainly work to dissipate profits, they are seldom as strong as in the model of perfect competition. In fact, aside from certain financial, agricultural, and mineral markets, few industries exhibit anything approaching perfect competition. Perfect competition is important for strategy not because it is common, but because departures from it underlie high profits. It establishes a baseline for an important question: why don't imitation and entry dissipate all profits in real industries? In this section, we explore how violations of the assumptions of perfect competition can lead to profitability.

Few Participants. Real markets are often dominated by a handful of sellers or buyers that have considerable influence over market prices. The four largest makers of manmade fibers in the United States, for instance, served 98% of the market in 1992; the four largest manufacturers of household washers and dryers, 94%; and the four largest cigarette makers, 93%.⁴ Influential buyers are also common. The Big Three U.S. auto makers account for a vast portion of the market for certain auto components in the United States.

Firms that are price makers rather than price takers can influence market prices in order to generate profits. Take, as an extreme case, a *monopolist* who is a sole seller to a large number of buyers. The monopolist could, like a perfectly competitive firm, set price equal to its minimum average cost; however, it would earn no profit (above the cost of capital). By raising its price above the competitive level, the monopolist can generate substantial profits as DeBeers has done in the diamond trade. In

⁴ U.S. Census of Manufactures, 1992.

doing so, it increases its profit per unit and reduces the number of customers willing to buy the good at the higher price.⁵

A similar situation can arise with a dominant buyer, who forces a price down. As the price falls, fewer sellers are willing to supply the good in question, but the profitability of the buying firm increases.⁶

Another similar dynamic arises when sellers (or buyers) *collude*—that is, band together in cartels to hold up (or force down) prices. Consider, for instance, OPEC's longstanding efforts to hold up high oil prices. As OPEC's experience illustrates, cartels are fragile because sellers and buyers are constantly tempted to cheat on the very coalitions they form. For instance, once a cartel of sellers raises prices above the competitive level, cartel members are tempted to undercut the cartel price ever so slightly in order to gain market share. Such undercutting may continue until the industry is right back at the perfectly competitive outcome. In addition, antitrust laws in most countries forbid sellers from making formal agreements to collude. Unspoken, tacit agreements that result in collusive behaviors, however, do arise in some settings—for instance, in some stable, established markets with a clear market leader.

Barriers to Entry and Exit. Under perfect competition, firms can enter the market with no disadvantage relative to established players and can exit readily. This assures that firms flood into the market if prices rise above costs, knowing that they can exit if prices fall later.

In reality, of course, established firms often do enjoy certain advantages that new entrants cannot match easily. For instance, they may enjoy the benefits of prior production experience, they may have already established full production scale, and they may have good reputations with customers. Such barriers have two effects that take us away from perfect competition. First, they allow established firms to raise prices above *their* costs without attracting entrants, who suffer from higher costs or cannot generate equivalent buyer value. Second, barriers to entry keep the number of sellers small, and this has all the effects discussed above. In extreme cases, barriers to entry may leave only one firm serving a market.

Likewise, exit is far from free in most markets. Tangible costs of exit can be steep. Even in the United States, where it is relatively easy to dismiss employees, severance costs of \$50,000 to \$100,000 per worker are common. Beyond concrete costs, there are substantial intangible costs associated with retreat from a market. For instance, a firm's employees may be demoralized and its reputation tarnished by an exit. When a market is growing, exit costs may be unimportant, but markets that have too much capacity (due to declining demand or bad luck) and also have high exit barriers tend to be very unattractive places to do business. High exit costs permit a very different kind of departure from the perfect competition model: prices may fall below not just average, but marginal costs, and firms may lose money indefinitely. Firms might prefer to stay in business and suffer small losses rather than shut down and bear substantial exit costs.

Differences Among Products. Perfect competition is so cutthroat in part because buyers have the same willingness to pay for the products of all firms, implying that they switch instantly to the seller that offers the best price. This is clearly one of the most strained of the model's assumptions. Goods in actual markets are typically quite varied. Goods differ not only in their physical

⁵ Note that this is harmful from the point of view of society. As it raises prices above the competitive level, the monopolist pushes some buyers out of the market; they are no longer willing to buy at the higher price. As a result, there emerge some buyers who are not served even though their willingness to pay for the good is greater than the cost it would take to serve them.

⁶ The starting point for this particular dynamic must be a price above the perfectly competitive equilibrium. If we begin at the competitive equilibrium and force prices down, all sellers become unprofitable and exit the market.

characteristics, but also in the brand images they carry, the distribution channels they exploit, the information they embody, and so forth. The variety of shampoos at the local supermarket or the diversity of automobile models should reinforce this point. Location can be a particularly potent source of heterogeneity. The McDonald's outlets in Seattle and Miami may offer virtually identical physical products, but because of the distance between them, they are hardly close substitutes for one another.

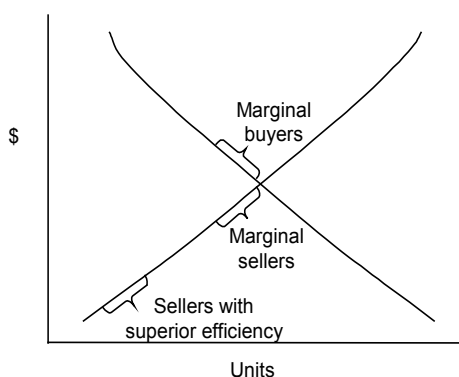
Product differences are an important departure from perfect competition. Once products in a market are not perfect substitutes for one another, firms can raise prices without losing all of their customers. In essence, each firm then exercises some limited monopoly power over the niche that it commands.

Differences Among Sellers. In the model of perfect competition, firms have equal access to technologies and inputs (or equivalently, the ability to imitate the most efficient firm costlessly). For that reason, they have identical cost structures. The assumption is crucial for the dire consequences of perfect competition. It assures that when the marginal firm, the firm that teeters on the brink of entry, earns zero profits, so do all of the firms that are active in the market.

In reality, of course, some firms have superior access to technology or inputs that give them lower costs or higher willingness to pay. Saudi Arabian oil producers, for instance, can extract purer oil more cheaply than many of their rivals. Because of its experience and scale, Intel can design microprocessors more effectively and produce them more efficiently than many of its competitors.

Firms with superior access to technologies or inputs can generate high profits even when the marginal firm earns nothing. Recall from Section II that the supply curve can be thought of as arraying firms or plants from most efficient to least. In markets that are not perfectly competitive, equilibrium is determined by the conditions of the incremental sellers and buyers (**Figure 14**). The price dictated by these marginal players may be high enough to make the most effective competitors very profitable. In the titanium dioxide industry, for example, the market price is determined by the production costs of old plants employing outdated sulfate technology; newer plants employing a more efficient chloride process have much lower costs and reap significant profits.

Figure 14 Equilibrium Determined by Marginal Buyers and Sellers



Imperfect Information. Full information about market conditions is important in the model of perfect competition. It ensures that buyers spot sellers who offer better prices and that potential sellers see when entry is profitable. In reality, of course, market information is far from readily available. Rather, buyers and sellers must spend time and money to gather and interpret market intelligence. This means, for instance, that sellers whose products constitute a small portion of buyers' total expenditures may be shielded from competition; buyers simply don't bother to search for better

vendors. It also bodes well for a supplier whose product's reliability is crucial for buyers' operations. Buyers are unlikely to leave an established, dependable supplier of a linchpin product for a new, unknown entrant. For example, appliance manufacturers have been slow to switch from steel sheet produced by integrated steel mills to sheet produced by minimills. Steel affects the quality and production of an appliance in crucial ways, and the appliance makers fear that the minimill product might foul up their stamping and finishing processes.

Imperfect information is closely related to two other forms of market imperfection. When information is costly to obtain and process, each buyer is likely to possess unique information about different sellers and their products. From the point of view of the individual buyer, then, this creates a situation in which sellers and their products look far from the same. In addition, buyers are likely to possess better information about established players than about new entrants. This can pose a barrier to entry. To enter a market effectively, a new firm might have to engage in marketing or advertising that established sellers do not require.

Analyzing Real Markets

Real markets differ in the degree to which they violate the assumptions of perfect competition and offer opportunities for profits. Tools such as the Five Forces framework help strategists to distinguish systematically between highly competitive markets and those in which rivalry is muted. When mastering these tools, it is worthwhile to consider how they relate to the model of perfect competition. How does each factor which makes an industry structurally attractive cause some departure from the rigors of perfect competition?

Finally, note that firms often have the capacity to create or reinforce departures from perfect competition. In the market for used cars, for instance, firms such as Republic Industries and Carmax are trying to consolidate a fragmented, highly competitive industry into one with a few major sellers. By offering a huge selection of cars, certifying the quality of vehicles, and adopting a no-haggle policy, these firms are attempting to distinguish their offering from that of the typical used-car outlet. The sophisticated information technology and extensive purchasing infrastructures they are putting in place are likely both to make entry more expensive than it has been in the past and to make such "mega-dealers" more efficient than the typical used-car dealer. The challenges involved in such an effort to mold industry structure are great, but so too are the potential payoffs.

Appendix: Social Efficiency of Equilibrium

Markets are socially valuable because they allow buyers and sellers to make mutually beneficial trades. The market for personal computers, for instance, might bring together a buyer who is willing to pay \$1,500 for a new computer and a seller that is willing to provide a computer for \$800. If this pair of buyer and seller consummates a trade, we say that the exchange has created $\$1,500 - \$800 = \$700$ of social value. The price of the computer, which could range from \$800 to \$1,500, determines which party captures the value. Regardless of the price, however, the trade creates a total of \$700 of value for society as a whole.

The total social value created by a market is simply the sum of the value created by each of the trades that occur in the market. Graphically, this is equal to the total area below the willingness-to-pay (or demand) curve and above the marginal cost (or supply).⁷

This note has discussed two types of market outcomes: the market equilibrium and its more extreme relative, the perfectly competitive equilibrium. The concept of competitive equilibrium relies on identical products, numerous small buyers and sellers, and full information. *Perfect* competition adds to these three features identical sellers and frictionless entry and exit. Economists often consider competitive equilibria and perfectly competitive equilibria to be virtuous because they generate, in certain senses, the maximal amount of total social value.

First consider a market equilibrium. (We discuss a perfectly competitive equilibrium below.) At equilibrium, all buyers and suppliers to the left of the equilibrium quantity engage in a trade. Since the willingness of each of these buyers to pay is greater than the marginal costs of each of these suppliers, all of these trades create social value.

To the right of the equilibrium point are buyers with willingness to pay below the market price. These potential customers opt not to buy at equilibrium. Also to the right are sellers who, because of their high marginal costs, are not willing to produce at the market price. A trade between one of these buyers and one of these sellers would *destroy* social value; production of the good would consume resources worth more than the final good is worth to the buyer.

You might initially think that a deal between the highest-value buyer and the just-excluded seller would create social value. Actually, it would not. Such a deal would leave a more efficient producer idle, and more resources than absolutely required would be used to serve the buyer.

In sum, competitive equilibrium generates the greatest possible social value. No good deal, which would boost total social value, goes undone. And no bad deal, which would destroy total social value, does get done. Put differently, the “right” buyers and sellers participate in the market at equilibrium. Because the market allocates goods “properly” – from the producers with the greatest efficiency to the buyers with the greatest desire – a market at its equilibrium is said to display *allocative efficiency*.

Market equilibria also exhibit the *law of one price*. No seller can hope to increase its profit by demanding a higher price, because the “marginal seller” – the seller with costs just slightly too high to make a sale in equilibrium – stands ready to undercut any price higher than the equilibrium price. Similarly, no buyer can hope to demand a lower price, since the seller simply turns to the just-excluded buyer, who is willing to pay virtually the market price. As a result, all trades take place at the market price.

The model of *perfect* competition requires more stringent assumptions than the simple competitive model; this results in a correspondingly stronger degree of social efficiency. In particular, perfect competition requires free entry and exit of equally efficient firms. As we discussed above, this results in a horizontal supply curve at the *minimum average cost*. All production in a perfectly competitive equilibrium, then, takes place in the most cost-efficient manner possible. This attractive feature is termed *productive efficiency*. Perfectly competitive equilibria also exhibit the allocative efficiency that is characteristic of all market equilibria.

⁷ This simplified analysis assumes that there are no fixed costs. More precisely, it considers the social value created in the market in the short-run since, if fixed costs are not avoidable in the short-run, they are not relevant to the short-run value creation associated with the equilibrium.

Caveats

Shallow students of economics learn about allocative and productive efficiency and conclude that unfettered competition is socially good at all times and in all places. This conclusion is unwarranted, for at least four reasons.

First, recall the stringent assumptions that underpinned the model of perfect competition. Real markets—in which products may be heterogeneous, buyers or sellers few in number, information imperfect, sellers heterogeneous, or entry and exit difficult—may not tend naturally toward allocative or productive efficiency.

Second, efficiency is by no means the same as equity. Perfect competition ensures that the most efficient producers provide goods to the buyers who desire those goods the most. It does nothing, however, to ensure that the resulting distribution of wealth is just in any ethical sense.

Third, the discussion throughout this note assumes that *all the costs and benefits of a transaction accrue directly to the seller and buyer involved in the transaction*. If costs or benefits accrue to parties external to the transaction, we say there are *externalities* associated with the trade. For example, the production of energy may provide benefits to the buyers who light their homes and power their factories, and may impose costs on the electric company in the form of fuel costs, wages, etc. But, the generation of energy may also generate pollution released into the air, which inflicts costs upon other consumers and businesses that were not party to the transaction. In this case, the total social value created by the transaction is not accurately reflected by the gap between the demand and supply curves, and the efficiency properties of the market equilibrium do not hold. There may also be so-called positive externalities in which non-participants to the transaction reap benefits, rather than costs. In general, a market equilibrium leads to too much production of goods that have negative externalities and to too little production of goods with positive externalities.

Finally, the traditional economic analysis of markets is static. Perfectly competitive markets give sellers strong incentives to produce efficiently at a point in time. They do not necessarily give firms adequate impetus to innovate and to upgrade their capabilities over time.

Acknowledgement

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